

# Overview of Markov decision process

*Subject: Markov decision process*

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## 1.

Continuous-time Markov decision process In discrete-time Markov Decision Processes, decisions are made at discrete time intervals. However, for continuous-time Markov decision processes, decisions can be made at any time the decision maker chooses. In comparison to discrete-time Markov decision processes, continuous-time Markov decision processes can better model the decision-making process for a system that has continuous dynamics, i.e., the system dynamics are defined by ordinary differential equations (ODEs). This modelling framework can be applied to areas such as queueing systems, epidemic processes, and population processes. Like the discrete-time Markov decision processes, in continuous-time Markov decision processes the agent aims to find the optimal policy that would maximize the expected cumulative reward. The key difference with the standard case is that, due to the continuous nature of the time variable, summation is replaced by an integral:

## 2.

Further reading Bellman., R. E. (2003) [1957]. Dynamic Programming (Dover paperback ed.). Princeton, NJ: Princeton University Press. ISBN 978-0-486-42809-3. Bertsekas, D. (1995). Dynamic Programming and Optimal Control. Vol. 2. MA: Athena. Derman, C. (1970). Finite state Markovian decision processes. Academic Press. Feinberg, E.A.; Shwartz, A., eds. (2002). Handbook of Markov Decision Processes. Boston, MA: Kluwer. ISBN 9781461508052. Guo, X.; Hernández-Lerma, O. (2009). Continuous-Time Markov Decision Processes. Stochastic Modelling and Applied Probability. Springer. ISBN 9783642025464. Meyn, S. P. (2007). Control Techniques for Complex Networks. Cambridge University Press. ISBN 978-0-521-88441-9. Archived from the original on 19 June 2010. Appendix contains abridged "Meyn & Tweedie". Archived from the original on 18 December 2012. Puterman., M. L. (1994). Markov Decision Processes. Wiley. Ross, S. M. (1983). Introduction to stochastic dynamic programming (PDF). Academic press. Archived from the original (PDF) on 2022-03-04. Retrieved 2019-01-19. Sutton, R. S.; Barto, A. G. (2017). Reinforcement Learning: An Introduction. Cambridge, MA: The MIT Press. Tijms., H.C. (2003). A First Course in Stochastic Models. Wiley. ISBN 9780470864289.

## 3.

There are multiple costs incurred after applying an action instead of one. CMDPs are solved with linear programs only, and dynamic programming does not work. The final policy depends on the starting state. The method of Lagrange multipliers applies to CMDPs. Many Lagrangian-based algorithms have been developed.

## 4.

Reinforcement learning is an interdisciplinary area of machine learning and optimal control that has, as main objective, finding an approximately optimal policy for MDPs where transition probabilities and rewards are unknown. Reinforcement learning can solve Markov-Decision processes without explicit specification of the transition probabilities which are instead needed to perform policy iteration. In this setting, transition probabilities and rewards must be learned from experience, i.e. by letting an agent

interact with the MDP for a given number of steps. Both on a theoretical and on a practical level, effort is put in maximizing the sample efficiency, i.e. minimizing the number of samples needed to learn a policy whose performance is  $\epsilon -$  close to the optimal one (due to the stochastic nature of the process, learning the optimal policy with a finite number of samples is, in general, impossible).

## 5.

Another application of MDP process in machine learning theory is called learning automata. This is also one type of reinforcement learning if the environment is stochastic. The first detail learning automata paper is surveyed by Narendra and Thathachar (1974), which were originally described explicitly as finite-state automata. Similar to reinforcement learning, a learning automata algorithm also has the advantage of solving the problem when probability or rewards are unknown. The difference between learning automata and Q-learning is that the former technique omits the memory of Q-values, but updates the action probability directly to find the learning result. Learning automata is a learning scheme with a rigorous proof of convergence. In learning automata theory, a stochastic automaton consists of:

## 6.

A Markov decision process (MDP) is a mathematical model for sequential decision making when outcomes are uncertain. It is a type of stochastic decision process, and is often solved using the methods of stochastic dynamic programming. Originating from operations research in the 1950s, MDPs have since gained recognition in a variety of fields, including ecology, economics, healthcare, telecommunications and reinforcement learning. Reinforcement learning utilizes the MDP framework to model the interaction between a learning agent and its environment. In this framework, the interaction is characterized by states, actions, and rewards. The MDP framework is designed to provide a simplified representation of key elements of artificial intelligence challenges. This modeling framework incorporates the understanding of cause and effect, the management of uncertainty and nondeterminism, and the pursuit of explicit goals. The name comes from its connection to Markov chains, a concept developed by the Russian mathematician Andrey Markov. The "Markov" in "Markov decision process" refers to the underlying structure of state transitions that still follow the Markov property. The process is called a "decision process" because it involves making decisions that influence these state transitions, extending the concept of a Markov chain into the realm of decision-making under uncertainty.

## 7.

where  $i$  is the iteration number. Value iteration starts at  $i = 0$  and  $V_0$  as a guess of the value function. It then iterates, repeatedly computing  $V_{i+1}$  for all states  $s$ , until  $V$  converges with the left-hand side equal to the right-hand side (which is the "Bellman equation" for this problem). Lloyd Shapley's 1953 paper on stochastic games included as a special case the value iteration method for MDPs, but this was recognized only later on. Value iteration is guaranteed to converge for  $\gamma < 1$  by the Banach fixed-point theorem.

## 8.

Discrete space: Linear programming formulation If the state space and action space are finite, we could use linear programming to find the optimal policy, which was one of the earliest approaches applied. Here we only consider the ergodic model, which means our continuous-time MDP becomes an ergodic continuous-time Markov chain under a stationary policy. Under this assumption, although the decision maker can make a decision at any time in the current state, there is no benefit in taking multiple actions. It is better to take an action only at the time when system is transitioning from the current state to another state. Under some conditions, if our optimal value function  $V^*$  is independent of state  $i$ , we will have the following inequality: